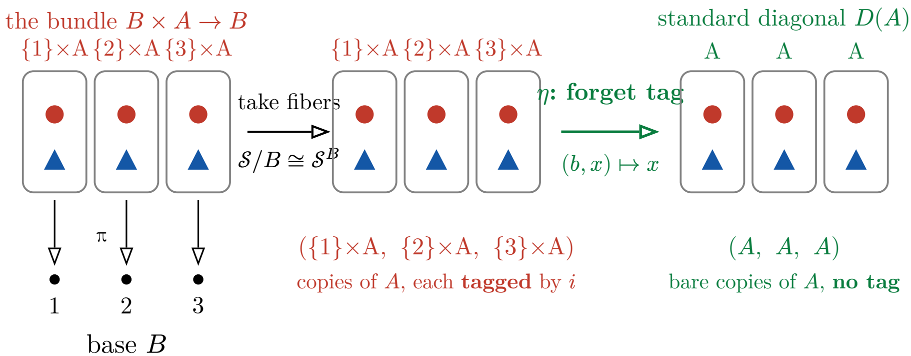
Why Freyd's Δ is conjugate to the standard diagonal

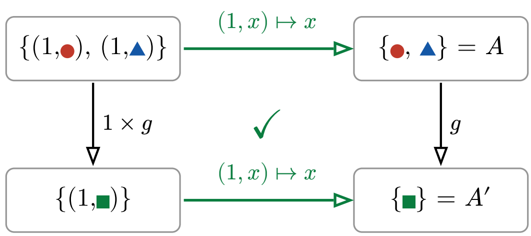
A set $B = \{1, 2, 3\}$, and a set A with two elements ● and ▲.



Groups 2 and 3 are the **same picture** — three copies of A — differing only by the header tag. The bijection that forgets the tag, $(b, x) \mapsto x$, is the isomorphism. That is all "conjugate" means here: Freyd's Δ lands on A **labelled by its base point**, the standard diagonal lands on A **plain**, and stripping the label is an iso.

...and the iso is natural — so it really is conjugate

Zoom into the fiber over 1. Hit A with any map $g : A \to A'$, say $A' = \{■\}$, g collapsing both ● ▲ to ■.



Both ways around send $(1, x) \mapsto gx$: Δ acts **inside** the fiber and never touches the tag 1, the diagonal acts in the slot — so forgetting-the-tag commutes with g. The same square holds over every base point. A tag-forgetting bijection that commutes with all maps **is** a natural isomorphism — i.e. Freyd's Δ is conjugate to the standard diagonal.

The four functors, precisely

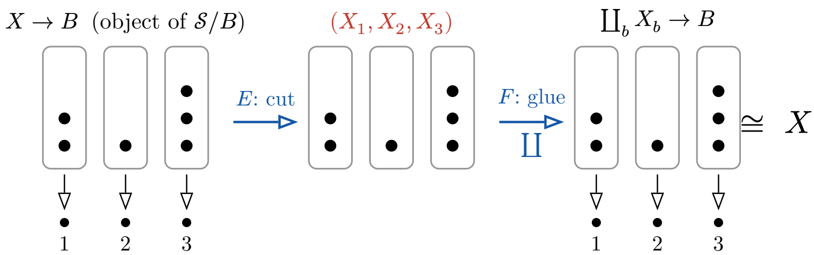
running example: $B = \{1,2,3\}$, and A any set

functor	**sends an object to**	**sends a map $g : A \to A'$ to**
D : standard diagonal $\mathcal{S} \to \mathcal{S}^B$	the constant family $(A)_{b\in B}$ i.e. (A, A, A)	the same g in every slot $(g)_{b\in B}$, i.e. (g, g, g)
Δ : Freyd's diagonal $\mathcal{S} \to \mathcal{S}/B$	the projection $\pi : B \times A \to B$	$1_B \times g$ (a map $B \times A \to B \times A'$ over B)
E : take fibers $\mathcal{S}/B \to \mathcal{S}^B$ the canonical equivalence	$(f : X \to B) \mapsto (f^{-1}(b))_{b\in B}$ the tuple of fibers	restrict $h : X \to X'$ to each fiber
F : glue $\mathcal{S}^B \to \mathcal{S}/B$ the inverse of E	$(X_b)_{b\in B} \mapsto \left(\coprod_b X_b \to B\right)$ disjoint union over B	act on each piece of the $\coprod$

Watch the types. D and Δ *both start at* $\mathcal{S}$ — $D : \mathcal{S} \to \mathcal{S}^B$ and $\Delta : \mathcal{S} \to \mathcal{S}/B$. They run in *parallel* they are not a forward/back pair. So *neither* $D \circ \Delta$ *nor* $\Delta \circ D$ is defined (each needs the other's target as its source, and both targets differ from $\mathcal{S}$). The single fact relating them is $E \circ \Delta \cong D$ — that is the entire top row of page 1.

The two composites that *are* Id (up to iso)

These are the round-trips of the equivalence $\mathcal{S}/B \cong \mathcal{S}^B$ — between E and F, *not* between D and Δ. Here a *general* bundle (fibers of sizes $2, 1, 3$):



$F \circ E \cong$ **Id** on $\mathcal{S}/B$ *(drawn)*: cut a bundle $X \to B$ into its fibers, then glue them back by disjoint union — you get $X \to B$ again.

$E \circ F \cong$ **Id** on $\mathcal{S}^B$: glue a family (X_b) into a total space, then re-cut into fibers — you get (X_b) again.

Each round-trip is the identity *only up to* the same tag-forgetting iso $(b, x) \mapsto x$: the glued set $\coprod_b f^{-1}(b)$ is the set of *pairs* (b, x), isomorphic but not equal to X. That "up to iso, not on the nose" is exactly why $\mathcal{S}/B$ and $\mathcal{S}^B$ are **equivalent** but not **isomorphic** categories — and Δ becoming D on page 1 is the same bookkeeping, restricted to the special bundles $B \times A \to B$.